

Name: _____

Exam 1 - MATH-262-02

Instructions: You have 110 minutes to take this exam. Please answer each question in the space provided. Answers without justification may **earn no credit**. You are permitted your signed prep work journal and a calculator on this exam. No other notes or electronic devices of any kind are allowed on this exam. Failure to comply may result in earning a 0 on any portion of this exam and/or being reported to the university.

Note: Unless otherwise stated, you may give implicit solutions.

1. [20 points] Classify the differential equation

$$x^2y'' + xy' + x^2y = y + \ln(x)$$

according to its type, order, linearity, and homogeneity.

2. [10 points] Verify that

$$r(\theta) = \frac{1}{\frac{1}{2} - \ln(\theta)}$$

is a solution to the equation

$$\frac{dr}{d\theta} = \frac{r^2}{\theta}$$

and find its largest interval of existence.

3. [18 points] Determine whether the differential equation

$$6y^2x \, dx - dy = 0$$

is separable, exact, and/or linear.

4. [14 points] Solve

$$2\frac{du}{dt} - u = 4\sin(3t).$$

Hint. $\int e^{ax} \sin(bx) \, dx = \frac{e^{ax}}{a^2+b^2} (a \sin(bx) - b \cos(bx)) + C.$

5. [16 points] Solve

$$\frac{2ty}{t^2 + 1} - 2t - (2 - \ln(t^2 + 1))y'(t) = 0.$$

6. [18 points] Use the substitution

$$u = 4x - y$$

to solve the IVP

$$y' = e^{4x-y}$$

While you may give your solution implicitly, it should not include u .

Hint. $\int \frac{1}{a-e^x} dx = \frac{1}{a}(x - \ln(a(a - e^x))) + C$.

7. [32 points] A large tank catches runoff from a chemical process. Suppose the tank initially contains 800 gallons of polluted water with 2 oz of pollutants. Polluted water runoff begins to flow into the tank at 3 gal/min. The runoff contains 5 oz of pollutants per gallon. The solution is piped out of the tank to a water treatment plant at 3 gal/min. You may assume the contents of the tank remain well-mixed.

(a) Write a first-order IVP that models the amount (in oz) of pollutants p after t minutes.

(b) Solve the equation you defined in part (a) using any method.

(c) How long does it take for the tank to accumulate 4 oz of pollutants?

(d) Suppose that regulation required the chemical process to be slowed if the tank ever contains more than 4oz of pollutants. When this happens, the runoff slows to 2 gal/min. How would this affect the ODE model you wrote in part (a)? The algebraic model you found in part (b)?